Magnetism in RealQM: Real Charge Currents, Diamagnetism, and the Geometry of Spin

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Abstract

RealQM formulates quantum mechanics as N real charge densities ψ_i^2 carried on non-overlapping domains $\Omega_i \subset \mathbb{R}^3$, minimising a global Coulomb energy with no self-interaction — the same three-dimensional continuum form as macroscopic mechanics, only smaller. We ask how much of magnetism follows from this real-charge ontology, and answer it in three graded steps. *Orbital magnetism* is native and requires no complex numbers: a magnetic moment is a real circulating current $\mathbf{j} = \rho\mathbf{v}$, carrying real angular momentum, with the complex wave function $\psi = \sqrt{\rho}\, e^{iS}$ merely bundling the two real fields $(\rho, \mathbf{v})$; the only irreducibly quantum input is the *integer* winding number, fixed by single-valuedness (quantised circulation), not by the complex plane. *Diamagnetism* gives a test against experiment: the Langevin susceptibility $\chi \propto -\langle r^2 \rangle$ is read directly from the RealQM density, and a radial self-consistent field reproduces the measured molar susceptibility of helium to $0.99\times$ and of neon and argon in size and trend ($\sim$20–30% too diamagnetic, the valence too diffuse without exchange), a diagnosed partial success. *Spin* is the single residue: Stern–Gerlach's two-spot splitting is pure two-valued spin-$\frac{1}{2}$ with $g = 2$, which the scalar theory does not carry. We show this residue is not what it is often taken to be. Its mechanical content is already real (a RealQM moment is a real current, so Einstein–de Haas is native, not an obstacle); its magnitude quantisation is topological closure (a real continuous configuration can only wind an integer, or spinorially a half-integer, number of times); and by Lévy-Leblond the factor $g = 2$ is a non-relativistic geometric fact (the SU(2) double cover), delivered for one electron by a $\boldsymbol{\sigma}\cdot\mathbf{p}$ structure with $g = 2$ emerging rather than posited. What remains genuinely open — the half-integer closure for the electron's own domain, and closed-shell pairing (which the spin-blind geometry does not supply) — is stated as such. The upshot is a clean ledger: magnetism's dynamics, energy and moment are real; spin is not a primitive but a single geometric factor to be derived; and where magnetism is real charge in three-dimensional space, RealQM captures it.

1 Introduction

Magnetism is a searching test of any ontology of matter, because it forces the question of what is really moving. In standard quantum mechanics the orbital moment is an operator $\hat{\mathbf{L}}$ acting on an amplitude, the associated "current" is a probability current physical only for one particle, and spin is an intrinsic two-valued property posited without a mechanical picture, its $g = 2$ arriving through the Dirac equation. RealQM takes a different starting point: the fundamental object is not an amplitude on $3N$-dimensional configuration space but N real charge densities in ordinary three-dimensional space [1]. The question of this paper is how much of magnetism follows from that commitment, and the answer is graded — solid for orbital magnetism, testable and partly confirmed for diamagnetism, and reduced to a single geometric residue for spin.

We proceed in that order. Section 2 recalls the RealQM setting and its current. Section 3 shows that orbital magnetism is real continuum mechanics with the complex form as mere bookkeeping. Section 4 confronts the diamagnetic susceptibility of closed-shell atoms with experiment. Section 5 isolates spin as a single geometric residue — Stern–Gerlach, the topological origin of its quantised magnitude, Einstein–de Haas as native, and $g = 2$ as non-relativistic geometry — and is explicit about what is delivered and what remains open. Section 6 states the ledger, and Section 7 the conditional thesis.

2 RealQM and its current

RealQM represents the electrons of an atom or molecule by N real, non-negative amplitudes ψ_i, each supported on its own region $\Omega_i \subset \mathbb{R}^3$, the regions non-overlapping, with charge density $\rho_i = q_i\psi_i^2$. The ground state minimises the global Coulomb energy — kernel attraction plus inter-electron and kernel repulsion — with *no electron self-repulsion*, the kinetic energy taking the confining role, and the free boundaries between domains located by continuity and a natural (Neumann) condition. Non-overlap plays the role that antisymmetry plays in the standard theory: it reproduces the spatial consequences of the Pauli principle — shells, the periodic table, bonding — with no spin variable. This is the same 3D continuum form as macroscopic mechanics, an ångström across.

Magnetism requires motion, and motion enters through a complex, time-dependent extension. Writing $\psi_i = R_i e^{iS_i}$ (Madelung form [2]) gives the charge density $\rho_i = q_i R_i^2$ and the current

$$\mathbf{j}_i = \frac{q_i}{m_i}\,\mathrm{Im}(\psi_i^*\nabla\psi_i) = \rho_i\mathbf{v}_i, \qquad \mathbf{v}_i = \frac{\nabla S_i}{m_i}, \tag{1}$$

so the phase gradient is the velocity of the charge, and $\mathbf{j}_i = 0$ for real ψ_i: a static real density carries no current. Because every ψ_i lives in real $\mathbb{R}^3$, each $(\rho_i, \mathbf{v}_i, \mathbf{j}_i)$ is a genuine three-dimensional charge fluid — one physical current per electron, for any N.

3 Orbital magnetism is real, and needs no complex plane

3.1 A velocity, not an i

It is often said that magnetism forces the theory complex: a circulating current needs a phase, and a phase means a complex wave function. The truth is sharper. From the density *alone* — a single real field $\psi = R$ — no magnetism can come, since $\mathbf{j} = \mathrm{Im}(\psi^*\nabla\psi) \equiv 0$: a lone amplitude carries no motion. But complex numbers are not the only way to carry a second field. Continuum mechanics never uses them, yet always carries *two* real fields, a density ρ and a velocity $\mathbf{v}$. Carried the same way here, the whole of orbital magnetism is real: the current $\mathbf{j} = \rho\mathbf{v}$ of (1) is a real circulating flow, costing a real kinetic energy

$$T_{\text{flow}} = \int \tfrac{1}{2}\rho|\mathbf{v}|^2\, d^3x = \int \frac{|\mathbf{j}|^2}{2\rho}\, d^3x, \tag{2}$$

carrying real angular momentum $L_z = \int \rho\,(\mathbf{x}\times\mathbf{v})_z\, d^3x$ and the ordinary magnetostatic moment $\boldsymbol{\mu} = \frac{1}{2}\int \mathbf{x}\times\mathbf{j}\, d^3x$; the magnetic state is the stationary point of the static energy augmented by (2) at fixed L_z (or, in a field, of $E - \boldsymbol{\mu}\cdot\mathbf{B}$). No i enters. The complex $\psi = \sqrt{\rho}\, e^{iS}$ is exactly these two real fields repackaged into one symbol, the phase being the velocity potential; the "i" is bookkeeping, and the physics is the flow.

3.2 Quantised circulation, and the moment

A winding state $\psi = R(\mathbf{x})e^{im\phi}$ carries an azimuthal current and the moment

$$\mu_z = -m\,\mu_B, \qquad m \in \mathbb{Z}, \tag{3}$$

quantised in the winding number and independent of the density's shape (the moment-arm $\propto s$ and the winding velocity $\propto 1/s$ cancel, the gyromagnetic relation $\boldsymbol{\mu} = \frac{q}{2m_e}\mathbf{L}$). The integer is the one genuinely quantum input, and it is topological, not a feature of the complex plane: single-valuedness of e^{iS} forces the circulation to close,

$$\oint \mathbf{v}\cdot d\boldsymbol{\ell} = \frac{2\pi m}{m_e}, \qquad m \in \mathbb{Z}, \tag{4}$$

a fractional winding being a torn, discontinuous configuration that a real continuous charge fluid cannot be. This is exactly the Onsager–Feynman quantisation of vortices in a real superfluid [3, 4]: one cannot make a half-strength vortex, because the flow must match up after a loop. What is unavoidable is the *circle* — the periodicity of an angle — not the complex plane. The density also vanishes on the axis for $m \neq 0$ (a hollow vortex core), since $\mathbf{v} \sim 1/s$ would otherwise diverge.

3.3 Numerical check: the Zeeman response

A Cartesian $(u, v) = (\operatorname{Re}\psi, \operatorname{Im}\psi)$ leapfrog for one electron in a two-dimensional harmonic well, initialised in the $m = 1$ state $\psi = (x + iy)e^{-r^2/2}$, conserves norm to 1.000000 and holds $\mu_z/\mu_B = -1.00$ (grid value -0.997) over thousands of steps: the circulating density is genuine. Adding a uniform field by minimal coupling $\mathbf{p} \to \mathbf{p} - q\mathbf{A}$ gives the Zeeman response of Table 1: the $m = \pm 1$ states split by $2\mu_B B$ to four digits, while the non-circulating $m = 0$ state is inert to linear order. Ordinary orbital magnetism, from a charge density in real space, with no spin invoked.

B	$E(m{=}0)$	$E(+1) - E(-1)$	$2\mu_B B$
0.05	0.999	+0.0499	0.0500
0.10	1.000	+0.0999	0.1000
0.20	1.004	+0.1998	0.2000

Table 1: Zeeman response of the circulating density (minimal coupling, atomic units). The $m = \pm 1$ splitting matches $2\mu_B B$; the $m = 0$ state is inert to linear order.

4 Diamagnetism: a test against experiment

A magnetic prediction can be read straight from the ground-state density, with no field applied. The Langevin–Larmor molar diamagnetic susceptibility is

$$\chi_M = -\frac{N_A e^2}{6 m_e c^2} \sum_i \langle r_i^2 \rangle = -0.792 \times 10^{-6} \Big(\textstyle\sum_i \langle r_i^2 \rangle / a_0^2 \Big)\ \mathrm{cm^3/mol}, \tag{5}$$

a sum over electrons of the mean-square radius — precisely the real-space quantity RealQM supplies as $\langle r^2 \rangle = \int r^2 \rho\, d^3x$. We evaluate it for the closed-shell atoms He, Ne, Ar from a radial RealQM self-consistent field: each electron moves in the kernel $-Z/r$ plus the Hartree field of *all the other* electrons, with no self-repulsion (the RealQM rule) and no exchange; the result feeds (5) directly (Table 2).

For **helium the prediction is essentially exact** (0.99×): the two-electron no-self-interaction field gives $\langle r^2 \rangle = 1.19\, a_0^2$ per $1s$ electron, agreeing with Hartree–Fock and with

atom	$\sum\langle r^2\rangle$ (a_0^2)	χ_{calc}	χ_{exp}	ratio
He	2.38	−1.88	−1.90	0.99
Ne	11.14	−8.82	−7.20	1.23
Ar	31.66	−25.1	−19.6	1.28

Table 2: Molar diamagnetic susceptibility (10^{-6} cm^3/mol) from the RealQM radial SCF via (5), against tabulated experiment.

the measured susceptibility — a magnetic observable *computed* from a RealQM density, not asserted. For **neon and argon the size and trend are right but the atoms are ∼20–30% too diamagnetic**; the per-shell breakdown pins the cause on the valence ($2p$ for Ne, $3p$ for Ar) coming out too diffuse. Two caveats frame this. First, the computation is a *radial mean field*: it enforces the no-self-interaction rule but spherically averages over the non-overlap geometry and omits exchange, both of which would *contract* the valence — so this is closer to a lower bound on the full three-dimensional solver than to its verdict. Second, the trend — excellent for the two-electron atom, degrading for the many-electron ones absent the full geometry — is the regime boundary RealQM shows elsewhere. It is a genuine partial success: one magnetic number secured against experiment, the right order and trend for the others, and the missing physics named.

5 Spin: the single geometric residue

5.1 What "spin" bundles

Two logically distinct things travel under the name spin. The first is fermionic *exclusion*, which RealQM realises geometrically by non-overlapping domains with no spin variable, and which accounts for shells, the periodic table, and bonding. The second is the two-valued *magnetic moment*, $g = 2$, which the scalar theory does not carry. This leaves exactly one residue — the electron's spin moment has $g = 2$, twice the orbital value per unit angular momentum — whose experimental face is Stern–Gerlach.

5.2 Stern–Gerlach: two with no middle, and the spread

A beam of silver atoms — each with a single valence electron in an $L = 0$ (orbitally spherical) state — passes through an inhomogeneous field and splits into *two* spots. This is the signature of a two-valued moment $\mu_z = \pm\mu_B$, i.e. spin $\frac{1}{2}$ with $g = 2$. An orbital moment would split into an *odd* number $2l + 1$, and for $L = 0$ silver would not split at all — the scalar winding ladder $m = 0, \pm1, \dots$ always *includes* zero, giving an undeflected middle beam. Stern–Gerlach gives two spots *with no middle*: the missing zero is the half-integer structure $m_s = \pm\frac{1}{2}$, which a scalar density cannot produce.

The *separation* of the spots, as against their number, is ordinary magnetostatics and native to RealQM. An atom of mass M and moment μ_z in a gradient $G = \partial_z B_z$ deflects by $z = \mu_z GL(D + \frac{1}{2}L)/Mv^2$ across a magnet of length L and drift D at speed v, so the split is

$$\Delta z = \frac{2\mu_B\, G\, L\, (D + \frac{1}{2}L)}{Mv^2}, \tag{6}$$

linear in the forcing G and $\propto 1/Mv^2$. The forcing sets the pattern's *scale*, continuously; it never changes the count (two) or the value ($\pm\mu_B$), and never fills the middle — stronger forcing only marches the same two dots apart, making the absent middle *more* conspicuous. The experiment thus factors a continuous classical magnifier (the deflection, native to a real moment in a real gradient) from a discrete quantised input ($\pm\mu_B$).

5.3 The mechanical content is already real (Einstein–de Haas)

One might fear a deeper obstacle: that spin's mechanical angular momentum, the $\hbar/2$ made visible by the Einstein–de Haas effect (a suspended bar physically rotating as it is magnetised and its moments align), is a further ingredient the theory must supply. In RealQM it is not. A magnetic moment here *is* a real circulating charge, and a real current carries real mechanical angular momentum by construction — the same L_z that accompanies the orbital $\mu_z = -m\mu_B$. So the reciprocal transfer of angular momentum between the aligned moments and the body — Einstein–de Haas, and its converse the Barnett effect — is automatic, ordinary conservation for a charge in motion, requiring no spin primitive. Everything *mechanical* about spin is already real in RealQM.

5.4 The quantised magnitude is topological closure

Why is the moment quantised to $\pm\mu_B$ rather than arbitrary? For the same reason the orbital moment is: a real, continuous, single-valued configuration can only wind an integer number of times ((4)); a fractional moment would be a torn configuration, which does not exist. The two attractors of any "steering" picture are therefore not free — they *are* the topologically allowed windings, and the continuum between them is not merely unstable but disallowed. For spin the same closure applies to a *spinorial* orientation rather than a scalar phase: a spinor closes only after 4π (the SU(2) double cover), so its winding is quantised in *half*-integers, giving the two-valued $\mu_z = \pm\mu_B$ with $g = 2$. Quantised magnitude is thus not an extra postulate but the price of the reality and continuity of the charge.

5.5 $g = 2$ is geometric, not relativistic

It is tempting to conclude that $g = 2$ forces relativity, since in the textbook account it falls out of the Dirac equation. By Lévy-Leblond's theorem [5] it does not: $g = 2$ follows from a first-order (linearised), *non-relativistic* Galilean spinor equation. Writing the kinetic term as $(\boldsymbol{\sigma}\cdot\mathbf{p})^2/2m$ — legal since $(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = p^2$ — minimal coupling gives

$$\frac{(\boldsymbol{\sigma}\cdot(\mathbf{p} - q\mathbf{A}))^2}{2m} = \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} - \frac{q}{2m}\,\boldsymbol{\sigma}\cdot\mathbf{B}, \tag{7}$$

the last term being exactly the $g = 2$ spin–Zeeman coupling, appearing automatically. The "2" is the SU(2) double cover of the rotation group — a spinor rotates at half angle, reciprocally doubling the moment — a geometric, scale-free fact, owing nothing to Lorentz invariance.

5.6 What is delivered, and what remains open

The single-electron step has been carried out numerically. On a two-dimensional grid a two-component spinor evolved with the $\boldsymbol{\sigma}\cdot\mathbf{D}$ structure ($\mathbf{D} = \mathbf{p} - q\mathbf{A}$) reproduces the operator identity $(\boldsymbol{\sigma}\cdot\mathbf{D})^2 = \mathbf{D}^2\nVdash - q\,\boldsymbol{\sigma}\cdot\mathbf{B}$ to grid accuracy, so the $g = 2$ term *emerges* from squaring rather than being inserted; and an $L = 0$ state — the very one a scalar density leaves undeflected — splits into two levels $\pm\mu_B B$ with no middle. That is Stern–Gerlach, non-relativistically.

We state plainly what remains a target. *(i) The half-integer closure* — that the electron's own domain carries a spinorial (4π) rather than scalar (2π) orientation — is exhibited by the $\boldsymbol{\sigma}\cdot\mathbf{p}$ structure but not yet *forced* by the domain geometry. *(ii) Closed-shell pairing.* A filled shell must pair to zero moment (diamagnetism), but the spin-blind non-overlap geometry does not supply antiparallel pairing: a naive per-electron spinor leaves the two spins degenerate and aligns them with any field (paramagnetic), and forcing continuity of the spinor across a shared boundary costs a spin domain-wall energy that *favours* parallel, the opposite of what is needed. Closed-shell magnetism therefore requires an added exchange term; it is a genuine limitation of the spin-blind geometry. Reproducing spin is thus complete for one electron and open for many —

and the measurement statistics across incompatible Stern–Gerlach axes (the non-commutativity revealed by sequential analysers) is the deeper frontier beyond a single splitting.

6 What is new, what is not, and the boundary

Because much of the phenomenology is standard, it is worth stating plainly what RealQM does and does not add.

New is the ontology. Magnetism becomes a literal charge fluid $(\rho, \mathbf{v})$ circulating in ordinary space — one genuine current per electron for any N — rather than an operator on a configuration-space amplitude; the complex form is bookkeeping; only the integer (orbital) or half-integer (spin) winding is quantum; and the two are located precisely against radiation (a temporal oscillation that genuinely needs the complex time-dependent form) as a *spatial* winding.

Not new is the phenomenology. RealQM *reproduces*, it does not rewrite, the gyromagnetic relation $\boldsymbol{\mu} = \frac{q}{2m_e}\mathbf{L}$, quantised circulation, the Zeeman splitting, and Langevin diamagnetism $\chi \propto -\langle r^2 \rangle$. The hydrodynamic $(\rho, \mathbf{v})$ form is Madelung's [2]; quantised vortices are Onsager's and Feynman's [3, 4]. The novelty is in *what these quantities are*, not in their values.

The boundary is collective magnetism. Exchange-driven magnetism — singlet–triplet splitting, ferromagnetic ordering — requires spin exchange that geometric non-overlap does not deliver, as the closed-shell test above makes concrete. Single-electron spin is reached as a residue; the collective sector is not, and no perspective here resolves it.

7 Conclusion

Magnetism in RealQM comes in three registers. Orbital magnetism is real continuum mechanics: a charge vortex with moment, angular momentum, and Zeeman response, the complex plane a convenience and the integer winding the only quantum. Diamagnetism is testable and, for helium, tested and matched; for heavier closed shells the size and trend are right and the shortfall diagnosed. Spin is a single geometric residue — not a mechanical mystery (Einstein–de Haas is native), not an arbitrary magnitude (topological closure), not a relativistic quantity ($g = 2$ is SU(2) geometry) — delivered for one electron and open for many. Nothing called spin need be *posited*: its exclusion role is geometry, its angular momentum is real charge motion, and what remains is a single geometric factor to be *derived*. The honest thesis is a conditional under adjudication: *where magnetism is real charge in three-dimensional space, RealQM captures it*, and the open frontier is to measure how far that reaches — phenomenon by phenomenon, extending the picture where charge alone is not enough, and marking honestly where it meets a limit.

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